ER Diagram

* What is ER - Diagram?

-> ER Diagram stands for Entity Relationship diagram.

-> Entity Relation model is a model for identifying entities (like student, car or company) to be represented in the database and representation of how those entities are related.

-> The ER Data model specifies enterprise schema that represents the overall logical structure of a database graphically.

-> Gather the requirements (functional and data) by asking question to ~~be~~ the database user.

-> Do a logical or conceptual design of the database.

-> This is where ER model plays a role.

-> It is the most use used graphical representation of how those entities are related. ~~The ER data model~~

-> Physical Database Design (like indexing) and external design (like views)

-> It is the most used graphical representation of the conceptual design of a database.

* Why we use ER Diagram

→ ER Diagram represents the ER-model in a database, making them easy to convert into relations (tables)

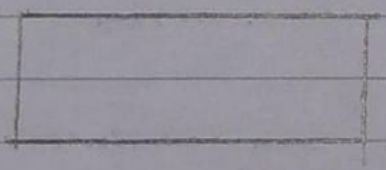
→ ER Diagram provides the purpose of real world modeling of objects which makes them intently useful.

→ ER Diagram provide no technical knowledge of a underlying DBMS used.

→ It gives a standard solution for visualizing the data logically

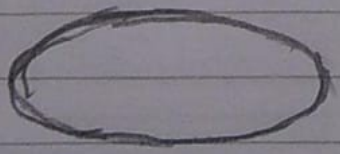
* Symbol of ER Diagram.

1) Rectangle



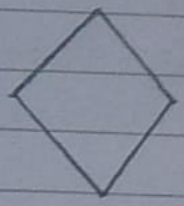
Rectangle represents Entity in the ER model

2) Ellips



Ellips represents Attributes in the ER model

3) Diamond



→ Diamond represents Relationship among Entities.

4 Line

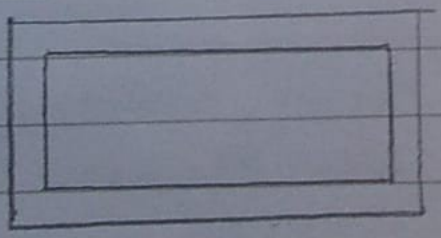
→ Line represents attributes to entity and entity sets with other relationship types.

5 Double ellips



→ Double ellipes represent multivalued Attribute.

6 Double Rectangle



→ Double Rectangle represents a weak entity.

CLASS Diagram

* What is class Diagram

A UML class diagram is a visual tool that represents the structure of a system by showing its classes, attributes, method, and the relationship between them.

→ It helps everyone involved in a project - like developers and designer.

→ Class Diagram are a type of UML (Unified modeling language) diagram used in software engineering to visually represents the structure and relationship of classes within a system i.e., used to construct and visualize object-oriented system

→ Class diagram provide a high-level overview of a systems design helping a communicate the structure of the software.

→ They are fundamental tool in object oriented design and play a crucial role in the software development life cycle.

→ In object-oriented Program (OOP), a class is blue print of object.

→ object are instance of a class.

→ Each class Define a set of attribute (data member)

→ The attribute represent the characteristics or properties of object, while the method define the behaviors or actions that the object can perform.

Why we use a class diagram :-

→ UML class diagram is essential tool in software engineering for several reasons.

→ Firstly, they help model a system class structure by representing classes and their attributes, method, and relationship.

→ This provides a clear and organised view of the system's architecture, making it easier to understand and communicat the design

→ Secondly they help facilitate understanding relationship between different part of the systems.

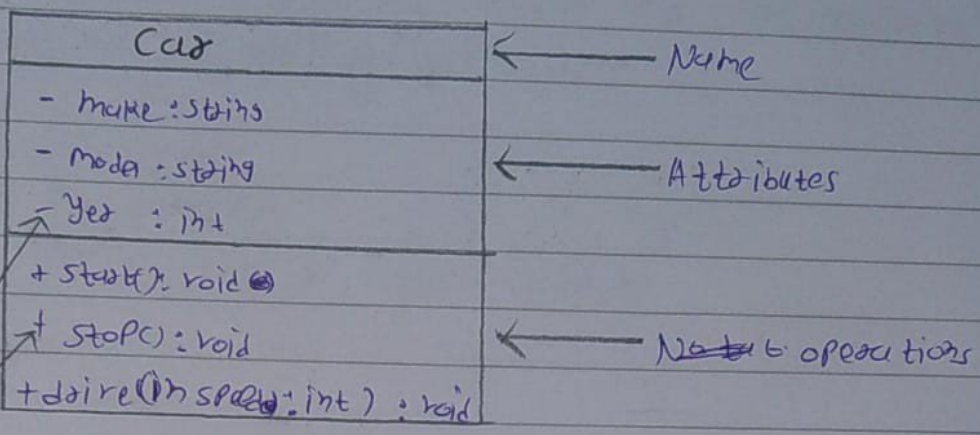
→ These UML Usecase diagram and UML class Diagrams depict associations, aggregations, compositions, inheritance and dependencies, helping stake holder understand how various system components are interconnected.

→ Thirdly, they provide a visual and standardised representation that can be easily interpreted by both technical and non technical audience, bridging the gape between different perspective.

→ Another reason to use UML Diagram is code generation.

→ Some software development tools support generating code directly from class diagram, which significantly reduces manual error and saves valuable development time.

* Symbol



1) Class Name

- The Name of the class is typically written in the top compartment of the class box and in centered and bold.

2) ~~Method~~ Attributes

→ Attributes also known as properties of field, represents the data members of the class.

→ They are listed in the second compartment of the class box and often include the visibility (e.g. Public, Private) and the data type of each attributes.

3)

3) operations

operations is also known as method, represent behavior or functionality of the class. They are listed in the third compartment of the class. box include the visibility return type, and parameters of each method.

Visibility Notation

Visibility Notations indicate the access level of attributes and methods, common visibility notation includes.

+ for Public
- for Private
for Protected
~ for Package or default visibility.